Inverse-Root Support and Fourier Fusion for Polynomial Exponential Periods

All-degree low-rank coefficient atlas and rank-preserving extensions

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Abstract

For a complex polynomial phase (P), the cyclic rank generated by the fixed seed (dx) can be read from the nonzero residue channels of the normalized inverse series at infinity, equivalently from the reduced constant span of the generic inverse roots. Earlier work established this bridge, a Fourier law for block fusion under polynomial decomposition, and a finite coefficient atlas. This successor gives an all-degree classification for cyclic rank at most three, relative to named results on primitive polynomial monodromy and finite projective linear groups. It proves that the only families, up to affine equivalence and lower-rank intersections, are powers, Dickson polynomials, quadratic- and cubic-over-power phases, indecomposable quartics, and the degree-twelve quartic collision. It also classifies every proper rank-preserving outer extension: powers extend by powers, Dickson polynomials extend within one dihedral tower, an odd-centred cubic-over-power phase admits one aligned square, and the exceptional quartic admits one cubic collision. The proof replaces an analytic boundedness step by algebraic invariant-integral descent, supplies the missing full-block stabiliser interface, repairs a false uniform Kummer-kernel assertion by a coefficient-stratified theorem, and handles arbitrary mixed Dickson chains by a direct rational-function invariant-field argument. Lean and two separate CAS implementations verify selected local consequences; two separate internal specialist re-audits pass. The result remains an unrefereed theorem candidate, not a complete formalisation, external reproduction, peer-reviewed theorem, or priority claim.

Keywords: polynomial exponential periods; inverse-root support; cyclic rank; polynomial decomposition; Ritt moves; Dickson polynomials; Kummer transport.

中文摘要

對任意複係數多項式相位 (P)，固定種子 (dx) 所生成的循環秩，可由無窮遠處規範化反函數級數中非零的同餘通道讀出；等價地，它也是一般反根之約化常數線性張成空間的維數。前一版本已建立此一橋樑、給出多項式分解下的 Fourier 區塊融合公式，並完成有限次數的係數圖譜。本後繼版本在若干具名的本原多項式單值群與有限射影線性群分類定理之條件下，提出循環秩不超過三的全次數分類。除去低秩交集並容許仿射等價後，候選族恰為冪多項式、Dickson 多項式、冪上的二次與三次相位、不可分解四次式，以及十二次四次碰撞。所有保持秩的非平凡外延亦被分類：冪塔、同一二面體塔中的 Dickson 外延、奇中心三次相位唯一的對齊平方，以及特殊四次式唯一的三次碰撞。證明以代數的「不變量—整性」下降取代解析有界性，補足完整區塊穩定子介面，分層修正先前錯誤的統一 Kummer 核主張，並以有理函數不變域直接處理任意混合 Dickson 鏈。Lean 與兩套分離的 CAS 實作只驗證局部推論；兩項分離的內部專業重審均通過。本成果仍是未經同儕審查的定理候選，並非完整形式化、外部獨立重現、已審定定理或優先權聲明。

關鍵詞： 多項式指數週期；反根支撐；循環秩；多項式分解；Ritt 變換；Dickson 多項式；Kummer 傳輸。

Candidate status and correction record

> **Version 0.6.0-candidate. Public unrefereed theorem candidate.** The mathematical gate passes internally relative to the classifications named in the proof. The release is not externally refereed, independently reproduced, completely formalised, or certified novel. Its novelty assessment is bounded by the openly documented search coverage.

This is an additive successor to version 0.5.0. During adversarial review, the claimed full Kummer kernel was found to be false on (B, A^2=4B) for even exponent. The release preserves that counterexample and replaces the uniform assertion with a coefficient-stratified theorem and an exceptional-stratum descent. A second gap concerning arbitrary mixed Dickson chains is closed by a direct invariant-field proof for the full outer polynomial. The two repairs passed separate internal specialist re-audits.

The historical v0.5.0 package and its manifest remain unchanged. This paper is assembled from this front matter, the sealed proof `ALL_DEGREE_GLOBAL_COMPLETION.md`, and `ALL_DEGREE_RELEASE_BACK_MATTER.md`.

Reproducibility and assurance map

The main replay command is:

```
python3 run_global_completion_checks.py
```

The reader-facing paper is rebuilt and checked for raw-TeX leakage with:

```
python3 tools/render_all_degree_paper.py
python3 tools/pdf_tex_preflight.py All-Degree-Low-Rank-Coefficient-Atlas.pdf
```

The v2 global receipt authenticates sixteen Lean theorems, ten exact global CAS checks, the repaired nineteen-check Kummer component, the direct Dickson proof, both internal re-audit reports, and thirteen adversarial receipt mutations. The bounded Dickson grid remains a regression control and is explicitly forbidden from carrying an all-degree claim.

The exact claim boundary, hashes, imported sources, negative controls, search coverage, and unresolved external-review obligations are recorded in `GLOBAL_COMPLETION_ASSURANCE.md`, `GLOBAL_COMPLETION_MANIFEST.sha256`, `programme/GLOBAL_COMPLETION_QUALITY_GATE.md`, and `programme/GLOBAL_NOVELTY_GATE.md`.

Global completion: root span, pole descent, primitive factors, and extensions

Status: **complete theorem candidate relative to named classification theorems**
Date: 2026-08-21

1. Purpose and assurance boundary

The preceding successor `ALL_DEGREE_STRUCTURAL_REPAIR.md` proves the monodromy/scaling lemma and the uniform all-`m` Newton obstruction. Four interfaces remained:

1. the inherited equality between inverse support and reduced root span;
2. the finite-pole and infinity-order descent used on powered elementary symmetric functions;
3. the low-dimensional primitive-factor audit; and
4. the classification of rank-preserving extensions.

This note closes those interfaces in the complex polynomial setting. The primitive and extension arguments import, with exact locators, the named classification of primitive polynomial monodromy and a current public preprint on composition-length-two kernels. Those classification theorems are not reproved here. The new local arguments, the source-to-case map, and the algebraic consequences are independently checkable within this package.

The result is therefore an internally complete all-degree theorem candidate, not an externally refereed theorem and not a formalisation of the imported classification of finite groups.

Throughout, `q(P)` denotes the fixed-seed cyclic rank, `W_P` the reduced constant span of the inverse roots, and `Sigma(P)` the nonzero inverse-at- infinity residue support after affine normalisation.

2. The inherited root-span interface

Let `P` be monic and centred of degree `d >= 2`. There is a unique series

$$\xi(t) = t + O(t^{-1}) \in t\,\mathbf{C}[[t^{-1}]] \quad \text{with} \quad P(\xi(t)) = t^d. \tag{2.1}$$

Write its residue decomposition

$$X_r(t) = \frac{1}{d}\sum_{j=0}^{d-1} \zeta_d^{-jr}\xi(\zeta_d^j t), \qquad \Sigma(P) = \{r \bmod d : X_r \ / \ 0\}. = \tag{2.2}$$

The inverse branches at infinity are

$$x_j = \xi(\zeta_d^j t) = \sum_{r \bmod d} \zeta_d^{jr} X_r. \tag{2.3}$$

Theorem 2.1 — Root span equals inverse support

For every complex polynomial `P`,

$$\boxed{\dim_{\mathbf{C}} W_P = |\Sigma(P)|.} \tag{2.4}$$

Proof

The matrices in (2.2) and (2.3) are inverse discrete Fourier matrices, so the constant span of the branches `x_j` equals the constant span of the channels `X_r`. Distinct nonzero channels are linearly independent over `C`: every Laurent exponent in `X_r` is congruent to `r` modulo `d`, and different residue classes are disjoint. Centring gives `X_0=0`, which is precisely the removed trace channel. This proves (2.4). The computation takes place in `C((t^{-1}))`; passing to the algebraic function field generated by the roots does not change a constant-linear dimension. □

The formal-moment isomorphism already proved in `MANUSCRIPT.md`, Theorems 4.3 and 5.2, identifies the active seed channels with the same set `Sigma(P)`. Consequently

$$\boxed{q(P) = |\Sigma(P)| = \dim_{\mathbf{C}} W_P.} \tag{2.5}$$

No analytic contour comparison is used in (2.4) or (2.5).

Corollary 2.2 — Right-factor projection

If `P=R circ Q`, `m=deg Q`, and `d=m n`, then

$$q(Q) = |\Sigma(P) \bmod m|. \tag{2.6}$$

Indeed, Fourier inversion within a `Q`-block groups the channels according to their residues modulo `m`. Channels with different residues have disjoint Laurent supports. This also proves heredity `q(Q)<=q(P)`. Equality holds if and only if the projection `Sigma(P) -> Z/mZ` is injective; equivalently every generic `Q`-block spans all of `W_P`. We call this **full fusion**.

This is the only root-span input used below.

3. Finite-pole and infinity-order descent

Let `A(Y)` be monic of degree `n`, let `y_1,...,y_n` be the roots of `A(Y)-z`, and choose elements `v_i` such that

$$v_i^m = y_i - b. \tag{3.1}$$

After the scaling lemma, monodromy acts by

$$(v_1, \dots, v_n) \longmapsto \epsilon_\gamma (v_{\sigma(1)}, \dots, v_{\sigma(n)}), \qquad \epsilon_\gamma^m = 1. \tag{3.2}$$

Put `f_k=e_k(v_1,...,v_n)`.

Lemma 3.1 — Invariant-integral descent

For every `1 <= k <= n`, `f_k^m` belongs to `C[z]`. Its pole order at infinity is at most `k/n`.

Proof

Equation (3.2) sends `f_k` to `epsilon_gamma^k f_k`, so `f_k^m` is fixed by the full geometric monodromy and therefore lies in the rational base field `C(z)`.

Each `y_i` is integral over `C[z]`, being a root of the monic polynomial `A(Y)-z`. Each `v_i` is integral over `C[z,y_i]` by (3.1). Transitivity of integrality makes every `f_k`, and hence `f_k^m`, integral over `C[z]`. Because `C[z]` is integrally closed in `C(z)`,

$$f_k^m \in C(z) \cap \overline{C[z]} = C[z]. \tag{3.3}$$

At a place above infinity, `y_i=O(z^{1/n})` and `v_i=O(z^{1/(mn)})`. Every summand of `f_k` is a product of `k` such terms, so

$$f_k^m = O(z^{k/n}). \tag{3.4}$$

This is the asserted infinity bound. □

Corollary 3.2 — Constancy below the top coefficient

If `k<n`, then `f_k^m` is a polynomial of integer degree at most `k/n<1`. It is therefore constant. Since `C` is algebraically closed and the function field is an integral domain, `f_k` itself is constant.

For `k=n`, there is an exact identity, including its leading coefficient:

$$f_n^m = \prod_i (y_i - b) = (-1)^n (A(b) - z). \tag{3.5}$$

Thus the final coefficient is affine and nonconstant. The varying radical polynomial consequently has all lower coefficients constant and only its constant term varying. This justifies the input `B(t)+e t^n` used in the uniform coefficient extraction, without an analytic boundedness assumption.

4. A stable line and the first rank increase

Lemma 4.1 — Stable line forces a power

If `W_P` contains a nonzero monodromy-stable line, then `P` is affine- equivalent to a power.

Proof

Let the line carry a character `chi`. At a chosen point over infinity, only the leading inertia character can have a pole. A stable line in any other character would have no pole over infinity and, because roots are integral over `C[z]`, no finite pole; it would be constant, impossible in the reduced span. Hence `chi(tau)=zeta_d` for the full `d`-cycle `tau` at infinity.

If H is a root stabiliser, the projection of that root to the line is nonzero and fixed by H, so H<=ker chi. Now [G:H]=d, while the image of chi contains an element of order d. Hence H=ker chi. The root action is faithful, so the core of H is trivial; normality gives H=1 and G=C_d. A complex polynomial with cyclic geometric monodromy is affine- equivalent to x^d. □

Therefore every nonpower root-span representation of dimension two or three is irreducible.

Choose a complete decomposition and the maximal power suffix x^m+b. Let A be the next indecomposable factor. Its generic roots determine distinct projective radical lines spanning a space of dimension at most three. The monodromy of A is primitive and contains a full cycle.

For completeness, the radical lines are distinct because a proportionality u_i=lambda u_j would imply y_i-b=lambda^m(y_j-b). Since a generic root is transcendental over C, this is the polynomial identity A(lambda^m(Y-b)+b)=A(Y). A nonidentity affine deck symmetry gives a nontrivial quotient decomposition of A; the remaining identity case would make two roots equal. The maximal-power choice excludes the sole indecomposable affine-deck case.

5. Source-exact primitive-factor audit

The audit uses two complete imported lists.

1. Theorem 3.3 of Behajaina–Koenig–Neftin, *Monodromy groups of polynomials of composition length 2*, arXiv:2603.27609v2, states that an indecomposable complex polynomial has monodromy of one of the following types: prime cyclic or dihedral; degree-four S_4; natural S_n or odd natural A_n; or one of the explicitly listed almost-simple groups in degrees 6,7,8,9,10,11,13,15,21,23,31.
2. The finite primitive subgroups of PGL_2(C) are A_4,S_4,A_5; the primitive subgroups of PGL_3(C) are A_5, PSL_2(7), A_6, H_1, H_2, H_3. The latter table is Table 2 of Calbeto, *On the finite irreducible subgroups of GL_n(C) and PGL_n(C)*, arXiv:2510.00718v1.

The older source Burda--Khovanskii, arXiv:1209.5137, gives the same Muller list in its theorem labelled mullerthm; it is retained as an independent source-to-list comparison.

Proposition 5.1 — Low-dimensional primitive audit

If the distinct radical lines belonging to the indecomposable factor A span dimension at most three, then A is affine-equivalent to

1. a prime power;
2. a prime Dickson/Chebyshev polynomial; or
3. a quartic with monodromy S_4.

Proof

In dimension two, three distinct projective points make the projective action faithful. Intersecting the PGL_2 list with the polynomial-monodromy list leaves the cyclic and dihedral families and the formal candidates S_4 in degree four and A_5 in degree five. Their point stabilisers are S_3 and A_4. A finite subgroup of PGL_2(C) fixing a point is cyclic, so neither candidate can stabilise a radical line. The solvable cases are precisely prime powers, prime Dickson polynomials, and degree four.

In dimension three, an irreducible imprimitive finite linear group is monomial: its diagonal subgroup is abelian and its quotient permutes three one-dimensional imprimitivity spaces through `S_3`. It is therefore solvable. A nonsolvable candidate must consequently act primitively. The radical lines are distinct (proportional radicals would give a constant affine relation between two distinct generic roots and hence a nontrivial block for the primitive root action), so the induced permutation action on those lines is the faithful root action. Thus every nonsolvable candidate embeds faithfully in `PGL_3(C)` and belongs to the primitive table quoted above.

The source-exact intersection of the two lists is now only

$$A_5 \text{ in degree } 5, \qquad PSL_3(2) \simeq PSL_2(7) \text{ in degree } 7. \tag{5.1}$$

For the three-dimensional `A_5` character and its point stabiliser `A_4`,

$$\langle \chi|_{A_4}, 1\rangle = \frac{3 + 3(-1) + 8(0)}{12} = 0. \tag{5.2}$$

The two nontrivial linear characters of `A_4` give the same zero because the three-cycle character value is zero. Hence `A_4` fixes no projective line.

For the Klein three-dimensional representation of `PSL_3(2)` and point stabiliser `S_4`, the character values on identity, order two, order three, and order four are `3,-1,0,1`. Therefore

$$\langle \chi|_{S_4}, 1\rangle = \frac{3 + 9(-1) + 8(0) + 6(1)}{24} = 0. \tag{5.3}$$

Multiplication by the sign character also gives zero:

$$\frac{3 + 6 - 3 - 6}{24} = 0. \tag{5.4}$$

Thus `S_4` fixes no projective line. Both exceptional candidates are excluded, proving the proposition. □

This expands every group, degree, stabiliser, and character used by the old one-paragraph audit. No unspecified “finite exceptional list” remains.

6. The full-block kernel lemma

Let `P=R circ Q`, and let `V_1,...,V_N` be the spans of the generic `Q`-blocks, where `N=deg R`.

Lemma 6.1 — A fixed full-span block detects the identity

Assume `q(P)=q(Q)`. Then `V_i=W_P` for every `i`. If an element of the block kernel fixes one `Q`-block pointwise, it is the identity.

Proof

Heredity gives `dim V_i=q(Q)=q(P)=dim W_P`, and `V_i<=W_P`; hence `V_i=W_P`. An element fixing the roots in one block fixes their span, so it acts identically on `W_P`. Every inverse root of `P` belongs to `W_P`; the element fixes all roots. The root action is faithful. □

Corollary 6.2 — Partial-support kernels are impossible

Inside the wreath-product embedding, every coordinate projection of the block kernel is injective. In particular, no nonidentity kernel element can be supported on a proper subset of the outer blocks.

This is the missing implication between full fusion and monodromy rigidity. It is stronger than the invalid shortcut “two tetrahedra span the same space, therefore they are the same tetrahedron”.

Lemma 6.3 — The full block stabiliser induces the absolute block monodromy

Let `M/C(t)` be the splitting field of `P(X)-t`, let `z` be one root of `R(Y)-t`, and let `B` be the corresponding block of roots of `Q(X)-z`. Then the setwise stabiliser `S_B` induces the full geometric monodromy group of `Q` on `B`.

Proof

The block system is the one attached to the intermediate field `C(t,z)`. Consequently

$$S_B = \mathrm{Gal}(M/\mathbf{C}(t,z)).$$

Let `L_B` be the splitting field of `Q(X)-z` over `C(z)`. It is an intermediate Galois field of `M/C(t,z)`. Restriction in a finite Galois extension is surjective onto the Galois group of every Galois intermediate field, so

$$S_B \twoheadrightarrow \mathrm{Gal}(L_B/\mathbf{C}(z)) = \mathrm{Mon}(Q).$$

If `g` carries `B` to another block `B'`, then `gS_Bg^{-1}=S_{B'}`. Under rank preservation, both block spans equal `W_P`; hence the linear action of `g` on `W_P` conjugates the two full induced block-monodromy representations. This is the full-group conjugacy used by the normaliser arguments below, rather than a claim that a coordinate image of the outer block kernel is automatically the absolute monodromy. □

7. Rank-preserving extensions in ranks one and two

Rank one

Lemma 4.1 gives `Q=x^m`; applying it to `P` gives

$$P = x^{mn}, \qquad R = y^n. \tag{7.1}$$

Dickson rank two

Uniformise a generic `D_m(x,a)` fibre by

$$x = U + a/U, \qquad W_Q = \langle U, U^{-1} \rangle. \tag{7.2}$$

If a second fibre with parameter `V` has the same plane, write

$$V = \alpha U + \beta U^{-1}, \qquad V^{-1} = \gamma U + \delta U^{-1}. \tag{7.3}$$

The identity `VV^{-1}=1` and nonconstancy of `U` give

$$\alpha\gamma = \beta\delta = 0. \tag{7.4}$$

Invertibility leaves `V=cU` or `V=cU^{-1}`. We now identify the entire extension, without a prime-factor or adjacency assumption.

Let `M/C(t)` be the splitting field of `P(X)-t`. Writing the roots in the reference `Q`-block as

$$x_j = \zeta_m^j U + a\zeta_m^{-j} U^{-1},$$

the two Fourier projections (valid for `m>=3`) are

$$U = \frac{1}{m}\sum_{j=0}^{m-1} \zeta_m^{-j} x_j, \qquad \frac{a}{U} = \frac{1}{m}\sum_{j=0}^{m-1} \zeta_m^{j} x_j.$$

Thus the reference block recovers `U` and `a/U`, so `C(U)<=M`. Rank preservation makes every block span equal to `<U,U^{-1}>`. Applying (7.3)–(7.4) to a uniformizer of every other block shows that every inverse root of `P` lies in `C(U)`. Hence

$$M = \mathbf{C}(U). \tag{7.5}$$

Every element of `Gamma=Gal(C(U)/C(t))` therefore sends `U` to `cU` or `c/U`. Thus `Gamma` is a finite subgroup of the normalizer of the multiplicative torus in `PGL_2(C)`. It contains the absolute block group

$$D_m = \langle U \mapsto \zeta_m U,\ U \mapsto a/U \rangle.$$

The rotational subgroup of `Gamma` is finite cyclic, say `mu_N`, and the displayed reflection gives `|Gamma|=2N`. On the other hand,

$$[\mathbf{C}(U) : \mathbf{C}(t)] = 2 \deg P = 2m \deg R = 2mn.$$

Thus `N=mn` (and in particular `mu_m<=mu_N`), and

$$\Gamma = \langle U \mapsto \zeta_N U,\ U \mapsto a/U \rangle \cong D_N. \tag{7.6}$$

Its invariant field is

$$\mathbf{C}(U)^{\Gamma} = \mathbf{C}\left(U^N + \frac{a^N}{U^N}\right) = \mathbf{C}(D_N(x,a)). \tag{7.7}$$

Since this field is `C(t)=C(P(x))`, the two generators differ by a Mobius transformation. Both are polynomials in `x` with their sole pole at infinity, so that transformation is affine. Consequently `P=A circ D_N(x,a)` for an affine `A`. Since `N=mn`, the Dickson identity gives

$$R \circ D_m = A \circ D_n \circ D_m.$$

Substitution by the nonconstant polynomial `D_m` is injective on `C[Y]`, so `R=A circ D_n`. The extension is exactly

$$D_{mn}(x,a) = D_n(D_m(x,a), a^m). \tag{7.8}$$

This argument classifies an arbitrary outer polynomial `R` at once. It therefore includes mixed aligned/nonaligned complete decompositions, repeated prime degrees, and every sequence of Ritt moves; no induction from a prime-degree chain theorem is required. A degree-two Dickson factor is affine-equivalent to a quadratic power and is absorbed by the power/Dickson overlap before the exact-rank-two argument is invoked.

Quadratic over a power

For `Q=(x^m+b)^2`, a block plane has a basis `U,V` satisfying

$$U^m + V^m = -2b \;/\; 0. \quad = \tag{7.9}$$

An equal block plane produces an invertible linear substitution preserving this level. Homogenising shows that it preserves the binary Fermat form `U^m+V^m`. For `m>=3`, its linear automorphism group is generated by coordinate scalings with `m`-th power one and interchange of the coordinates. Thus the new fibre parameter is the old one or its negative before squaring, which is the same `Q`-fibre. No distinct full-fusion block is created.

For `m=2`, this family is the degree-four Dickson overlap and is already covered by (7.8). Hence Dickson chains are the only proper rank-two- preserving extensions.

8. Rank-three cubic-over-power extensions

Write

$$Q = H(x^m), \qquad H(Y) = Y^3 + AY^2 + BY + C, qquadm \geq 2. \tag{8.1}$$

Exact rank three means `(A,B)!=(0,0)`, except that `m=2`, `B=A^2/4!=0` is the rank-two Dickson overlap. The affine-power locus `3B=A^2` remains rank three for `m>=2`, but its cubic monodromy is `C_3` and must be treated separately from the `S_3` branch. In a generic fibre, the three Kummer blocks span independent lines with generators `U_i`, where the powered values `Y_i=U_i^m` have fixed first and second elementary symmetric functions.

Lemma 8.1 — Kummer-line transport on the full-kernel strata

Off the exceptional stratum `B!=0`, `A^2=4B`, `2|m`, transport between two generic `Q`-blocks under a rank-preserving extension has the form

$$U_i' = c_i U_{\pi(i)}. \tag{8.2}$$

Proof

By Lemma 6.3, the full setwise stabiliser of every `Q`-block induces the absolute block monodromy. An element carrying one block to another conjugates these full actions on the common three-dimensional span.

Off the exceptional stratum `B!=0`, `A^2=4B`, `2|m`, the Kummer rotations form the full diagonal normal subgroup of the monomial block action. For `m>=3`, its three weight lines are the unique one-dimensional imprimitivity system and hence are preserved by every such coordinate identification. For `m=2`, the

projective block action is the octahedral S_4; its projective normaliser is itself and preserves the orbit of the three opposite-axis pairs, which are the same Kummer lines. Thus in the nonpower cases a block transport permutes the three weight lines, giving (8.2). On the affine-power locus 3B=A^2, the projective exponent-two group is tetrahedral A_4; its normaliser is generated by octahedral signed-monomial matrices and diagonal character intertwiners, so it preserves the same axes. The spectral argument handles this locus for m>=3.

□

Put a_i=c_i^m. The powered triples Y_i and a_iY_i have the same fixed first and second symmetric functions. On the nonpower S_3 locus, the constant-linear relation space of the generic cubic roots is zero when A!=0 and is the trace line when A=0. Consequently all a_i are one common scalar c. The fixed coefficients give

$$cA = A, \qquad c^2B = B. \tag{8.3}$$

If A!=0, then c=1. If A=0 and B!=0, the sole nontrivial possibility is c=-1. Hence the complete fibre-equivalence group is at most mu_2, and equality occurs precisely when the centred cubic is odd. The only proper extension is

$$Q = x^{3m} + bx^m, \qquad P = Q^2, \qquad b \neq 0. \tag{8.4}$$

On the affine-power locus, write the roots as Y_i=-a+omega^i w. Preservation of the first two elementary symmetric functions gives

$$\sum a_i = 3, \quad \sum a_i\omega^i = 0, \quad a_1^2 = a_2^2 = a_3^2 = 1.$$

Therefore every a_i=1, and this locus has no nontrivial fibre quotient.

It remains to treat the exceptional stratum excluded from Lemma 8.1. Write A=2b, B=b^2, and m=2k. Then

$$H(x^{2k}) - C = (x^{3k} + bx^k)^2.$$

For k=1 this is, up to target translation, D_6 and has rank two. For k>=2 it is already the aligned-square quotient of the nonexceptional phase F=x^{3k}+b x^k. The nonexceptional calculation just completed proves that the full rank-preserving fibre quotient of F has degree at most two. Any further nonlinear outer factor on F^2 would therefore give a quotient of F of degree greater than two, which is impossible. No transport assertion for the reduced Kummer kernel is needed on this stratum.

The extension (8.4) cannot repeat. After rewriting it as a cubic over x^{2m}, the outer cubic is

$$Y(Y+b)^2 = Y^3 + 2bY^2 + b^2Y, \tag{8.5}$$

which is the exceptional even stratum already isolated in the proof of Lemma 8.1. It is already the degree-two quotient of x^{3m}+b x^m; a further nonlinear quotient would compose to a rank-preserving quotient of that phase of degree greater than two, contradicting the complete nonexceptional fibre-equivalence calculation.

9. Rank-three quartic extensions

Let

$$Q = x^4 + Ax^2 + Bx, \qquad B \neq 0. \tag{9.1}$$

Its monodromy is `S_4` and its root span is the standard three-dimensional representation.

Take the rightmost indecomposable factor `G` of an outer extension `R`; then `G circ Q` is a right factor of `R circ Q`. Heredity forces

$$q(Q) \le q(G \circ Q) \le q(R \circ Q) = 3, \tag{9.2}$$

so `G circ Q` is already rank preserving.

The indecomposable-monodromy list and the length-two kernel theorems of Behajaina–Koenig–Neftin are exhaustive:

- if `G` is an `AGL_1` polynomial, their Theorem 2.3 says that the kernel contains `V_4^{deg G}` in every case except the Ritt collision `X^3 circ ((X^3+1)X)`;
- if `G` has `S_4` monodromy, their Theorem 2.4 contains `V_4^4`;
- if `G` is nonsolvable, their Theorem 2.7 contains `V_4^{deg G}`.

Every displayed product contains a nonidentity element supported on one outer block, contradicting Corollary 6.2. Therefore only the Ritt collision can preserve rank. In coefficient form it occurs exactly when `A=0`:

$$Q(cx) = cQ(x) \quad (c^3 = 1), \qquad P = Q^3 = (x^4 + Bx)^3. \tag{9.3}$$

There is no further extension. The projective monodromy of (9.3) is the tetrahedral `S_4`; its normaliser in `PGL_3(C)` is itself because every automorphism of `S_4` is inner. Thus any further full-fusion transport is, after a root permutation, scalar. Coefficient comparison in

$$(x^4 + Bx)^3 = x^{12} + 3Bx^9 + 3B^2x^6 + B^3x^3 \tag{9.4}$$

shows that a scalar semiequivariance has `c^3=1` and target multiplier one. It is already a deck transformation and creates no distinct fibre.

10. Completion of the first-increase cases

The primitive audit leaves prime powers, prime Dickson factors, and quartics. The power computation gives ranks two and three only for outer degrees two and three. The Dickson calculation gives the aligned half-angle identity in rank two; its only rank-three first increase is the cubic case already covered by the uniform Newton descent. The quartic case is exactly the uniform quartic descent.

By Corollary 3.2 the lower radical coefficients in those descents are constant, so the coefficient-extraction argument applies without an analytic qualification. `ALL_DEGREE_STRUCTURAL_REPAIR.md` then proves:

$$\text{cubic lift: } m = 2, \qquad \text{quartic lift: } m = 3 \text{ and } A = 0. \tag{10.1}$$

The latter is (9.3). The former is the aligned Dickson/power-square overlap.

11. All-degree theorem

Theorem 11.1 — Complex low-rank coefficient atlas

Let `P in C[x]` have degree at least two. Up to invertible affine changes of source and target, `q(P)<=3` if and only if `P` belongs to one of the following families, with the lower-rank intersections removed in the stated rank:

1. rank one: powers `x^d`;
2. rank two: Dickson polynomials `D_d(x,a)` and quadratic-over-power phases `x^{2m}+A x^m`;
3. rank three: cubic-over-power phases `x^{3m}+A x^{2m}+B x^m`, indecomposable quartics `x^4+A x^2+B x` with `B!=0`, and the collision `(x^4+a x)^3` with `a!=0`.

The proper rank-preserving extensions are exactly:

$$\begin{aligned} &\text{powers:} && x^m \mapsto x^{mn}, \\ &\text{Dickson:} && D_m(x,a) \mapsto D_{mn}(x,a), \\ &\text{aligned square:} && x^{3m}+bx^m \mapsto (x^{3m}+bx^m)^2, \\ &\text{quartic collision:} && x^4+ax \mapsto (x^4+ax)^3. \end{aligned} \tag{11.1}$$

Proof

Read ranks from the right along a complete decomposition. Rank-one suffixes combine to a power. The stable-line lemma makes every nonpower rank-two or rank-three span irreducible. The `[2,1]` support profile would give a common stable line, so rank cannot increase from two to three. Proposition 5.1 and the uniform descent classify the first strict increase above the maximal power suffix. Sections 7–9 classify every subsequent rank-preserving step and show that the two rank-three steps cannot repeat. These alternatives are exactly the displayed families. Direct inverse-support computations, already sealed in the coefficient atlas and degree-32 successor, establish existence and the stated lower-rank intersections. □

Combining Theorem 11.1 with (2.5) removes the last all-degree qualifier from the algebraic fixed-seed atlas.

12. What is and is not closed

Closed internally in this successor:

- the local root-span/support interface needed by the atlas;
- finite-pole descent and the exact infinity-degree bound;
- the complete source-to-group primitive audit;
- the missing full-block-kernel implication;
- the four rank-preserving extension mechanisms and non-repetition; and
- the resulting all-degree classification over `C`, relative to the named finite-group and polynomial-monodromy classification theorems.

Not claimed:

- a reproof of the classification of primitive polynomial monodromy groups;
- a reproof of the finite-subgroup classifications of `PGL_2` and `PGL_3`;
- formal verification of algebraic function-field integrality or those group classifications;
- external independent reproduction, specialist peer review, novelty, or publication; or
- Laurent-polynomial, positive-characteristic, or arbitrary-amplitude extensions.

Sources used for imported classification steps

- A. Behajaina, J. Koenig, and D. Neftin, *Monodromy groups of polynomials of composition length 2*, arXiv:2603.27609v2, especially Theorems 2.1–2.7, and 3.3. Theorem 6.1 is a contextual cross-check for nonaligned odd-prime chains, not a load-bearing input after the direct proof in Section 7. https://arxiv.org/abs/2603.27609
- Y. Burda and A. Khovanskii, *Polynomials invertible in k-radicals*, arXiv:1209.5137, especially the theorem labelled `mullerthm`. https://arxiv.org/abs/1209.5137
- G. G. Calbeto, *On the finite irreducible subgroups of GL_n(C) and PGL_n(C)*, arXiv:2510.00718v1, Tables 1 and 2. https://arxiv.org/abs/2510.00718
- M. E. Zieve and P. Muller, *On Ritt's polynomial decomposition theorems*, arXiv:0807.3578. https://arxiv.org/abs/0807.3578

Data and code availability

The complete candidate package, exact receipts, Lean sources, CAS programs, negative controls, bounded manifests, source audits, and internal review reports are available from the public GitHub repository and the exact-version Zenodo record. The release archive is deterministic and its SHA-256 identity is recorded in the archive receipt. No confidential data are used.

Limitations

The proof imports named classification results for primitive polynomial monodromy and finite projective linear groups. The Lean layer verifies selected arithmetic, spanning, coefficient, and character consequences; it does not formalise algebraic-function-field integrality, the direct Dickson invariant-field argument, Kummer theory, Ritt theory, or the imported classifications. The separate CAS implementations check encoded algebraic consequences, not the source-to-encoding bridge in its entirety. The two specialist re-audits are internal and model-assisted, not external peer review.

The literature search used zbMATH Open, complete retrieved OpenAlex citing sets for named seeds, OpenCitations and Crossref checks for stable DOI seeds, Semantic Scholar cross-checks, publisher records, and AMS Batch MR Lookup for bibliography identity. A dated machine-readable snapshot preserves the provider-specific counts and their disagreements. It was not an exhaustive topical

MathSciNet search, all-index citation-forward search, non-English full-text search, or survey of unpublished work. No absolute novelty or priority claim is made. Positive-characteristic, Laurent-polynomial, zero-cycle, and arbitrary-amplitude extensions remain outside the theorem.

Ethics declaration

This mathematical research uses no human participants, animals, personal data, or sensitive datasets. No ethics approval was required.

Author contributions (CRediT)

Anonymous: Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing—original draft, and Writing—review and editing. Ian Pitchford served as repository maintainer and Evidence Press publisher; this operational role is not scholarly authorship.

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